KAAN GÜLER

Computer Engineer · Autonomous Systems · Edge AI & Real-Time Perception
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PROFESSIONAL SUMMARY

M.Sc. Computer Science student at the Karlsruhe Institute of Technology (KIT) and Computer Engineering graduate (High Honors, TED University). Led a 5-member computer vision sub-team in a funded autonomous vehicle research project, taking real-time perception models from data pipelines to on-vehicle deployment on NVIDIA Jetson hardware. Writes C++ and Python for performance-sensitive systems, ships software end to end, and works with design documentation, code review and testing discipline in team settings. Applying to the KIRA Rocketry and Space Robotics teams, with a software focus on autonomous navigation, perception, and avionics or ground-control software.

TECHNICAL SKILLS

Perception & Autonomy: YOLOv8, DeepSORT, U-Net segmentation; real-time perception pipelines; dataset and training pipelines; experiment tracking and tuning (MLflow, Optuna).

Edge Computing: NVIDIA Jetson (Orin Nano), TensorRT optimization; C++ (FFmpeg, FFI integration into Node.js); parallel computing coursework.

Software & Systems: Python (primary), C++, TypeScript; async FastAPI; REST API design; PostgreSQL; Docker; GitHub Actions (CI/CD).

Applied AI: LangGraph multi-agent systems; Google Gemini (VertexAI), OpenAI and Anthropic APIs; tool calling; RAG and vector search (Qdrant).

Engineering Practice: design documentation, code review, unit testing, mentoring, cross-functional teamwork.

EXPERIENCE

Computer Vision Sub-Team Lead | Autonomous Vehicle (Funded Research), TED University | Sep 2024 - Jun 2025

- Led a 5-member cross-functional sub-team that owned the real-time perception stack of an autonomous scale vehicle: model development, data pipelines, on-vehicle integration, and design documentation.
- Deployed real-time models on NVIDIA Jetson Orin Nano hardware and built an end-to-end ML platform with MLflow and Optuna for experiment tracking and hyperparameter tuning.
- Set code-quality standards and mentored junior team members.

Computer Engineer Intern | HAVELSAN (Defense Industry) | Jul 2024 - Aug 2024

- Built unit-tested C++ modules integrated into a Node.js (Express.js, Vue.js) service via FFI, following defense-sector engineering standards with code review and documentation discipline.

AI / Backend Engineer (Freelance) | Remote, Türkiye | Jun 2025 - Present

- Built and shipped a consumer mobile app to the App Store with a Google Gemini multimodal vision feature (VertexAI on GCP) that describes users' photographs of historical artifacts, plus an in-app antiques marketplace.
- Engineered the TypeScript backend and REST endpoints behind the image pipeline and marketplace, handling response latency, token cost, and error states.
- Took the app through App Store review to launch, resolving security and release-readiness issues.

Computer Engineer Intern | Usta Bilgi Sistemleri | Aug 2024 - Sep 2024

- Built backend components for a corporate document-signing application (.NET, React, PostgreSQL); containerized runtimes with Docker for predictable deployment.

Computer Engineer Intern | Ziraat Teknoloji | Aug 2023 - Sep 2023

- Developed RESTful APIs and MVC architecture for regulated financial applications (ASP.NET Core); refactored SQL queries to reduce latency in an audit-driven environment.

SELECTED PROJECTS

Jobmaxxing (CVSend) | Python, async FastAPI, Gemini, Next.js, PostgreSQL, Docker | 2026

- Built and deployed a full-stack AI product (Render + Vercel) with JWT auth, CI, and Docker; live demo at jobmaxxing-orpin.vercel.app.
- Engineered reliability guardrails: per-call token cost, latency and time-to-first-token tracking; DB-backed response caching; automatic multi-model fallback on quota limits; strict JSON output contracts.

TestForge | Python, LangGraph, OpenAI API, Docker, GitHub Actions | Jan 2026 - Mar 2026

- Built a four-agent LangGraph system (writer, runner, analyzer, suggester) that generates, executes, and analyzes software tests, with self-correcting loops and failure classification.

EDUCATION

M.Sc. in Computer Science | Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany | 2026 - Present

B.Sc. in Computer Engineering | TED University, Ankara | Sep 2020 - Jun 2025

- GPA 3.60 / 4.00 (4.00 = maximum), High Honors. Senior thesis on U-Net image segmentation. Relevant coursework: Artificial Intelligence, Parallel Computing, Data Structures & Algorithms, Database Systems, Software Engineering.

ADDITIONAL

- **Teams & Mentoring:** TEDU Game Development Society Supervisory Board; TEDU IEEE Unity Mentor (8 teams). Unity / C# game projects.
- **Languages:** Turkish (native), English (professional proficiency), German (learning).